Assignment 1

System pro.—

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3. Write a sequence of instructions for SIC to ALPHA equal to the product of BETA and GAMMA. Assume that ALPHA, BETA and GAMMA are defined as in Fig.1.3(a).

```
LDA     BETA
MUL     GAMMA
STA     ALPHA

ALPHA   RESW  1
BETA    WORD  5
GAMMA   WORD  2
```

8. Write a program that adds up 100 words of array NUMS and stores the result to a memory location called SUM

```
LDX   ZERO
LDA   NUMS,X
ADD   SUM
STA   SUM
LDA   XREG
ADD   WUS
STA   XREG
LDX   XREG
COMP  WUSSUP
JLT   Loop
```

```
ZERO     WORD 0
SUM      RESW 1
WUS      WORD 3
WUSSUP   WORD 300
XREG     WORD 0
```